

Predicting COVID-19 Treatment Outcomes from Seven Days of P/F Ratio Measurements

Semester Assignment

Analysis & Modelling of Complex Data — Big Data

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Summary. This report analyses 622 intensive-care patients, each with fourteen P/F ratio measurements taken twice daily over seven days, and predicts whether the recorded outcome is *Improving* or *Deteriorating*. One record holding a single imputed value repeated across all fourteen timepoints was removed, and 1.1 % of readings were winsorised at Tukey fences computed on the log scale. Every timepoint separates the two groups after correction for multiple testing, and the effect size grows from Cohen's $d = 0.22$ on admission to 0.95 on the evening of day seven. Three classifiers reach 0.70 to 0.74 accuracy under nested cross-validation against a 0.50 baseline, and none is significantly better than the others once the comparison is made correctly. Two properties of the supplied data are documented and neither is benign: the outcome labels run opposite to clinical expectation across nine of the ten feature files, and the label column contains a changepoint near row 500. A scan over row offsets rules out misalignment as the explanation for either.

1 Introduction and dataset

This report recreates a COVID-19 study in which physiological measurements taken during emergency care are used to predict a patient’s treatment outcome. BT code 731058 assigns the P/F ratio ($\text{PaO}_2/\text{FiO}_2$, in kPa), the standard measure of oxygenation efficiency and the variable used to grade the severity of acute respiratory distress syndrome.

The dataset holds 622 patients, each with fourteen measurements taken twice daily over seven days, and one outcome label per patient. Neither file carries an identifier, so the only thing linking a measurement row to its outcome is position. The notebook therefore asserts that the row counts and indices match before concatenating and that the outcome column gains no missing values afterwards, since a silent misalignment would be invisible from that point on.

The outcome classes are exactly balanced, 311 *Improving* against 311 *Deteriorating*. This is what makes accuracy a usable metric here. The majority-class baseline is 50 %, so any score above it reflects signal rather than a guess at the larger class; on an imbalanced dataset a classifier could reach 90 % accuracy by always predicting the majority, and the number would mean nothing. Balance also removes any need for resampling or class weights, and stratified splitting is enough.

The data contains no missing values whatsoever. For clinical records that is implausible, and Section 2 finds a record that shows the data has been imputed upstream. Two further properties emerged during inspection, and both bear on how the results should be read.

1.1 Direction of the outcome labels

A higher P/F ratio means better oxygenation, and a falling P/F ratio is the defining feature of worsening acute respiratory distress. In this dataset the group labelled *Deteriorating* has the higher P/F ratio, 23.8 against 18.9 kPa, and the rising trend, +0.11 against −0.29 kPa per half-day. Level and direction of travel both point the wrong way.

Table 1 checks whether this is peculiar to the assigned variable by evaluating the nine other feature files against the same `outcome.csv`. Against identical labels the *Deteriorating* group has lower CRP, lower lactate, lower peak pressure and lower PaCO_2 , with higher oxygen saturation, higher pH, higher bicarbonate and higher platelets. Nine of the ten level differences are significant at $p < 0.001$ and eight of the ten trend differences at $p < 0.05$, all describing the “deteriorating” patients as the healthier and recovering ones.

One of the ten shows nothing at all: ventilatory ratio differs on neither level ($p = 0.266$) nor trend ($p = 0.281$), and its values cluster around 1300 for a quantity that should sit near 1–3, so that file appears mis-scaled. The other qualification concerns how much “nine of ten” is worth as evidence. These variables might have been so intercorrelated as to amount to one measurement counted repeatedly, but they are not: the mean absolute correlation between their per-patient means is only 0.21, and the effective number of independent tests implied by the eigenvalues of that correlation matrix is 8.0 out of ten. They carry largely separate information. What they do share is the cohort, since all ten are measured on the same 622 patients against the same label vector, so this is one finding observed ten ways rather than ten independent replications of it.

1.2 Structure in the row ordering

The obvious competing explanation is that `outcome.csv` is misaligned with the feature rows rather than mislabelled. A *random* permutation can be ruled out immediately, because it would make the labels independent of the measurements and destroy the signal rather than reverse it. That argument does not, however, rule out a structured misalignment, so the row order itself was tested.

The ordering is strongly non-random. A Wald–Wolfowitz runs test on the outcome column gives 241 runs against an expected 312 ± 12.5 , which with a continuity correction is $z = -5.66$ and $p = 1.5 \times 10^{-8}$. That structure is not spread through the file: restricting the same test to the first 500 rows removes it entirely, at $z = -0.44$ and $p = 0.66$, so that portion is in perfectly random order. What the global test

Table 1: Group means and mean within-patient linear trend (per half-day) for all ten feature files evaluated against the same `outcome.csv`, with Welch p -values for each difference. Trends are fitted after clipping each column at its 1st and 99th percentiles so a single record cannot dominate a slope. The final column states what the *Deteriorating* group’s values would ordinarily imply.

Variable	Mean			Slope			Reading of “Deteriorating”
	Impr.	Deter.	p	Impr.	Deter.	p	
CRP	272.55	220.04	<.001	−0.81	−3.65	.004	less inflammation
HCO ₃ [−]	27.31	28.37	<.001	+0.34	+0.34	.885	less acidotic
Lactate	1.58	1.35	<.001	+0.01	−0.00	.004	better perfusion
Oxygen saturation	93.48	94.08	<.001	−0.12	−0.05	<.001	better oxygenation
P/F ratio	18.92	23.82	<.001	−0.29	+0.11	<.001	better, improving
PaCO ₂	7.07	6.55	<.001	+0.08	+0.02	<.001	better ventilation
Peak pressure	27.85	24.95	<.001	+0.23	−0.08	<.001	better compliance
Platelet count	264.63	310.98	<.001	+3.24	+6.62	<.001	less coagulopathy
<i>Ventilatory ratio</i>	<i>1348.13</i>	<i>1260.77</i>	<i>.266</i>	+2.63	−16.77	<i>.281</i>	<i>no difference</i>
pH	7.351	7.392	<.001	+0.001	+0.003	<.001	less acidotic

detects is a single changepoint near row 500: the *Deteriorating* share is 59.6 % over rows 0–499 and 10.7 % over the remaining 122, and a χ^2 test of homogeneity across bins of a hundred gives $\chi^2(6) = 101.5$, $p = 1.2 \times 10^{-19}$.

The misalignment hypothesis can be tested directly. If the two files were offset by some number of rows, realigning them should strengthen the association between measurements and labels. Scanning offsets from −30 to +30 and testing the group difference in per-patient mean at each, the association is strongest at an offset of exactly zero ($t = +8.84$), the nearest competitor is less than half as strong ($t = +5.03$ at +3), and no offset in the range produces a significant association in the clinically expected direction, the most negative statistic being $t = -0.61$. Realigning the files never restores the expected direction, so misalignment is ruled out and mislabelling stands.

The ordering is therefore a second and separate defect rather than the cause of the first, namely a label file whose last fifth is almost entirely one class. It affects nothing downstream, since the split is stratified and shuffled, but every clinical statement in this report should be read with the caveat that the class named *Deteriorating* behaves throughout like a recovering patient.

2 Data cleaning

Cleaning proceeds in two tiers, an integrity check for values that are unambiguously wrong followed by a statistical rule for values that are merely unusual.

2.1 Integrity

There are no missing cells, no non-positive values and no duplicated patient rows. One record fails: patient 84 holds the identical value 22.166667 at all fourteen timepoints. Fourteen independent measurements taken over seven days cannot agree to six decimal places, and 22.166667 is the repeating decimal $66.5/3$, which is what a single value carried forward across a series looks like. The record contains no temporal information and would contribute a slope of exactly zero to every trajectory statistic. It was removed, leaving 621 patients. This is the only removal made on integrity grounds, and it is also the direct evidence that some upstream imputation took place.

2.2 Outliers

Tukey’s rule, flagging values outside $[Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}]$ column by column, marks 128 patients, 20.6 % of the dataset. Discarding a fifth of a clinical cohort is a large enough step to justify looking at the distribution first.

The variable is strongly right-skewed at every timepoint, with mean absolute skew 1.75 and a D’Agostino–Pearson test rejecting normality at $p \approx 3 \times 10^{-50}$. Tukey’s rule assumes rough symmetry, and applied to skewed data it flags the whole upper tail whether or not anything in it is wrong. Figure 2 shows the mechanism directly: the upper fence falls at roughly 40 kPa and cuts through a smoothly decaying tail, with no gap separating the body of the data from the supposed outliers. A log transform brings mean absolute skew down to 0.24 and leaves only marginal evidence against normality ($p \approx 0.03$).

Rather than adopt one rule and defend it afterwards, five candidates were compared on what they cost, what they do to the class balance, and what they do to downstream accuracy (Table 2).

Table 2: Five cleaning strategies. Accuracy is five-fold cross-validated SVC accuracy on the resulting dataset. The five span 0.018, against a fold-to-fold standard deviation of roughly 0.04, so this column cannot decide between them. Note also that the rows being scored differ between strategies, which makes the accuracies not strictly like for like.

Strategy	Removed	%	n left	Impr.	Deter.	SVC CV accuracy
Keep everything	0	0.0	621	311	310	0.726
Drop: IQR $k = 1.5$ (raw)	128	20.6	493	268	225	0.718
Drop: IQR $k = 3.0$ (raw)	40	6.4	581	296	285	0.733
Drop: IQR $k = 1.5$ (log)	57	9.2	564	291	273	0.736
Winsorise: IQR $k = 1.5$ (log)	0	0.0	621	311	310	0.723

Accuracy does not separate the strategies, and treating a gap of 0.01 as evidence would be reading noise. The strategies differ sharply in who gets deleted, and that is what settles the choice. The naive rule removes 85 *Deteriorating* patients against 43 *Improving*, close to two to one, because *Deteriorating* is the higher-valued class and Tukey’s rule has only an upper fence to trip. Applying it would turn a balanced dataset into a 268/225 imbalanced one while removing exactly the observations that carry the discriminative signal. The log-scale rule is gentler but not immune, at 37 against 20.

The primary dataset is therefore produced by winsorising at the log-scale fences. No patient is deleted beyond the integrity removal, the 311/310 balance survives, and the extreme readings are capped so they cannot dominate a mean, a variance or a distance-based classifier. A patient with one unusual reading on day three still contributes thirteen sound measurements, and deleting the record throws those away as well. In total 97 readings out of 8694, or 1.12 %, were capped across 57 patients.

Because the brief asks how many rows were removed, a second dataset was built in which those 57 rows, 9.2 % of the total, are deleted instead, leaving 564. The whole classification comparison is repeated on it in Section 6.4.

2.3 Limitations

Flagging column by column treats the fourteen timepoints as independent and ignores that each patient is a time series, so a value that is unremarkable in the marginal distribution but represents an impossible jump from the previous reading goes undetected; a longitudinal rule would be better but needs clinical thresholds that were not available. Winsorising compresses the tails, so the standard deviations in Section 3 are mildly optimistic. Patient 84 establishes that imputation happened upstream, so an unknown share of the capped readings may be artefacts rather than measurements. Finally the fences are fitted on the whole dataset including the eventual test split, where strictly they belong inside the resampling loop. That last one was measured rather than left as a worry: refitting the fences inside each cross-validation fold changes SVC accuracy from 0.7681 to 0.7679, so the leakage is real but negligible at this sample size.

3 Statistical analysis

3.1 Change over time

In aggregate both groups are close to flat (Table 3, Figure 1). The *Improving* group drifts down from 20.0 to 16.9 kPa over the week, a mean slope of -0.29 per half-day, and the *Deteriorating* group drifts up from 22.4 to 24.6 kPa at $+0.11$. Both within-group changes are significant under a paired Wilcoxon signed-rank test, at $p = 6.7 \times 10^{-11}$ and $p = 5.3 \times 10^{-6}$ respectively. Neither group undergoes a dramatic clinical course. What grows over the week is the distance between them.

Table 3: Group-level summary of the cleaned dataset ($n = 621$), mean (SD) in kPa unless stated. “Change” is the day-7 mean minus the day-1 mean, each averaged over that day’s two readings; “slope” is the least-squares trend per half-day across all fourteen timepoints. The last columns give Welch’s t for the between-group difference.

	Improving ($n = 311$)	Deteriorating ($n = 310$)	Welch t	p
Whole-series mean	18.87 (5.52)	23.70 (7.64)	—	$< 10^{-15}$
Day 1	20.04 (8.29)	22.40 (10.25)	3.16	0.0017
Day 7	16.95 (6.59)	24.65 (9.66)	11.58	$< 10^{-15}$
Change (day 7 – day 1)	−3.10 (7.84)	+2.24 (10.58)	7.15	2.7×10^{-12}
Slope (per half-day)	−0.292 (0.602)	+0.109 (0.773)	7.20	$< 10^{-11}$
Within-patient SD	5.05 (2.79)	6.02 (2.66)	4.44	$< 10^{-4}$

3.2 Differences between the groups

Each timepoint was tested with Welch’s t -test, appropriate because the group variances differ, alongside the Mann–Whitney U test, which assumes no distribution and is the safer of the two given the residual skew. Fourteen tests on one dataset call for a multiplicity correction, and Benjamini–Hochberg was used to control the false discovery rate. Its validity is worth one sentence rather than being taken for granted: BH requires independence or positive regression dependence, and the fourteen timepoints are *positively* correlated at every lag (Figure 4), which satisfies the second condition.

All fourteen timepoints remain significant after correction, which at $n = 621$ is close to unavoidable, so the effect size is the more informative quantity. Cohen’s d rises almost monotonically from 0.22 at day-1 morning, below the conventional 0.2 threshold for a small effect, to 0.95 at day-7 evening, a large effect and a 4.3-fold increase (Table 4). A single admission measurement carries almost no discriminative information; a week of them carries a great deal. The trajectory is separately informative, the day-7 minus day-1 change differing between groups at $p = 2.7 \times 10^{-12}$.

Table 4: Cohen’s d between the outcome groups at each timepoint, computed on the cleaned dataset. All fourteen are significant at a Benjamini–Hochberg-adjusted $\alpha = 0.05$. The value quoted is the equal- n form of the pooled standard deviation, which at 311 against 310 differs from the n -weighted form in the fourth decimal.

	D1 AM	D1 PM	D2 AM	D2 PM	D3 AM	D3 PM	D4 AM
d	0.22	0.26	0.38	0.39	0.51	0.51	0.55
	D4 PM	D5 AM	D5 PM	D6 AM	D6 PM	D7 AM	D7 PM
d	0.57	0.54	0.55	0.76	0.73	0.80	0.95

3.3 Correlation between timepoints

Adjacent timepoints correlate at $r = 0.713$, decaying monotonically to $r = 0.382$ across a full week, with a mean off-diagonal correlation of 0.560 (Figure 4). The fourteen columns are nothing like fourteen independent measurements; they are one smooth, autocorrelated trajectory. Two consequences follow

and both are acted on in Section 6. Multicollinearity makes regularisation matter for logistic regression, and the effective dimensionality is far below fourteen, which predicts that hand-built summary features should add nothing. That prediction was tested rather than assumed: adding the series mean, standard deviation, slope, day-7 minus day-1 change and the two day-level averages moves nested accuracy by +0.014, -0.007 and +0.001 for the three classifiers, which is no gain.

4 Data visualisation

Six figures follow, using five plot types: a line plot with confidence bands, a histogram with a density overlay, a grouped box plot, a heatmap and a scatter plot, closing with a panel of index plots. Five further figures supporting the classification appear in Section 6.

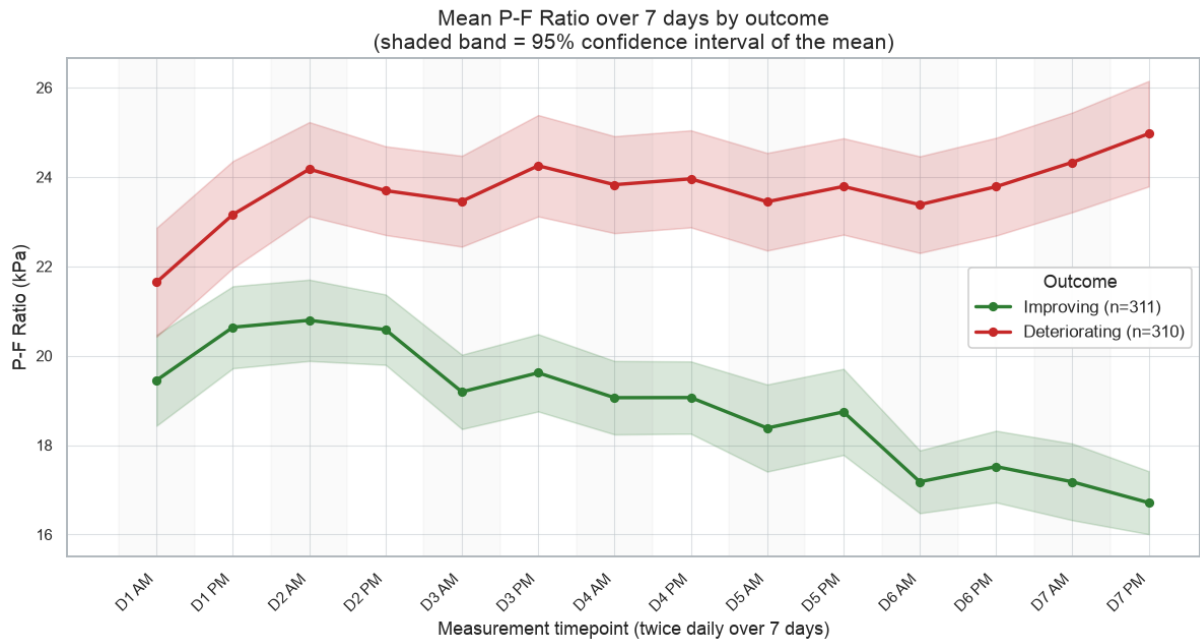


Figure 1: Mean P/F ratio at each of the fourteen timepoints by outcome. Shaded bands are 95 % confidence intervals of the mean and the alternating stripes mark the seven days. The gap widens from roughly 2.2 kPa on day one to 8.3 kPa on day-7 evening.

A line plot is the right choice because the fourteen columns are ordered timepoints rather than independent variables, and the confidence bands answer whether the groups are separable at each point rather than only on average. The gap widens monotonically: on day one the bands only just clear one another, a mean difference of about 2.2 kPa against a within-group standard deviation near 10, so the group means are distinguishable while individual patients are not. By day-7 evening they are apart by 8.3 kPa with a clear margin. The second point is that neither group moves far in absolute terms, so the separation comes from slow divergence rather than any clinical event. This is what predicts the modelling result in Section 6: a classifier given day-one data alone would perform near chance.

Figure 2 pairs a histogram with a kernel density estimate and draws the fences on top, which turns a distributional plot into direct evidence for a cleaning decision. The left panel shows a dense body between 10 and 30 kPa with a tail reaching 71 kPa, and an upper fence near 40 kPa passing through that tail with nothing on either side to suggest a separate population. The right panel is the same data after a log transform, close to symmetric at skew 0.25, with only a handful of values beyond the fences. This is the empirical basis for the decision taken in Section 2.

Group means can differ impressively while individual patients stay indistinguishable, so where Figure 1 compares averages, a box plot asks whether a single patient could be classified. Separation at day-7 evening is real and larger than the mean difference alone suggests: the *Deteriorating* median sits at the

Distribution of P-F Ratio at D1 AM, raw and log scale

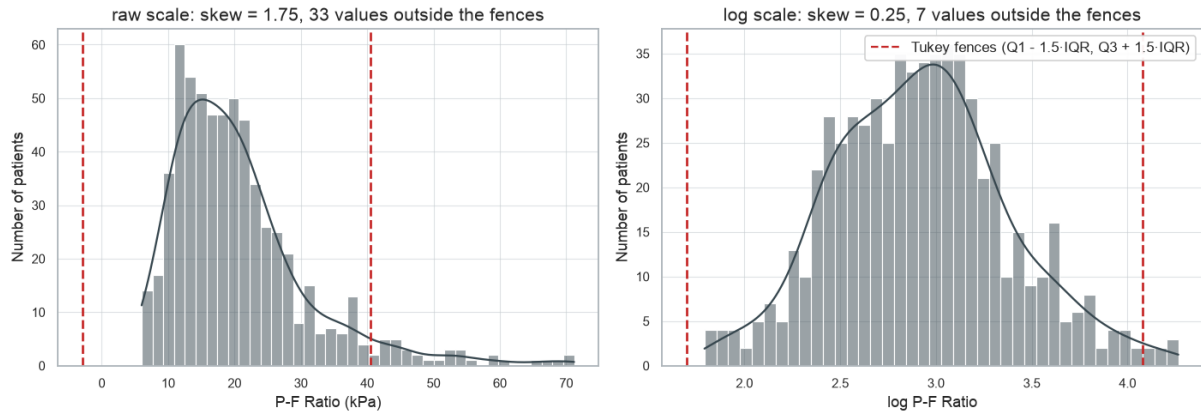


Figure 2: Distribution of the P/F ratio at day-1 morning on the raw scale (left) and the log scale (right), with the Tukey fences drawn in. On the raw scale the upper fence cuts through a smoothly decaying tail; on the log scale the distribution is near-symmetric and few values fall outside.

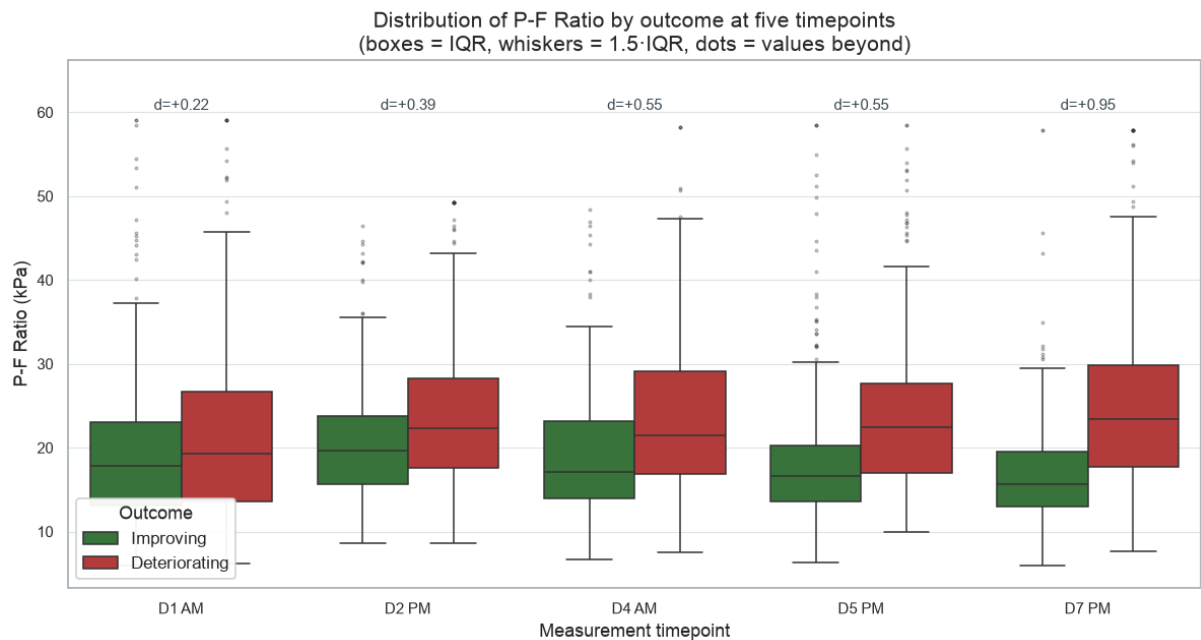


Figure 3: Distribution of the P/F ratio by outcome at five timepoints, with Cohen's d annotated above each pair. Even where the effect is largest the boxes overlap substantially.

Improving 91st percentile, so a threshold placed there would capture most of one group. At day-1 morning the same comparison gives the 58th percentile, which is barely better than chance. The residual overlap at every timepoint is the visual explanation for the accuracy ceiling in Section 6: a substantial minority of patients sit where no decision boundary can be right for both classes.

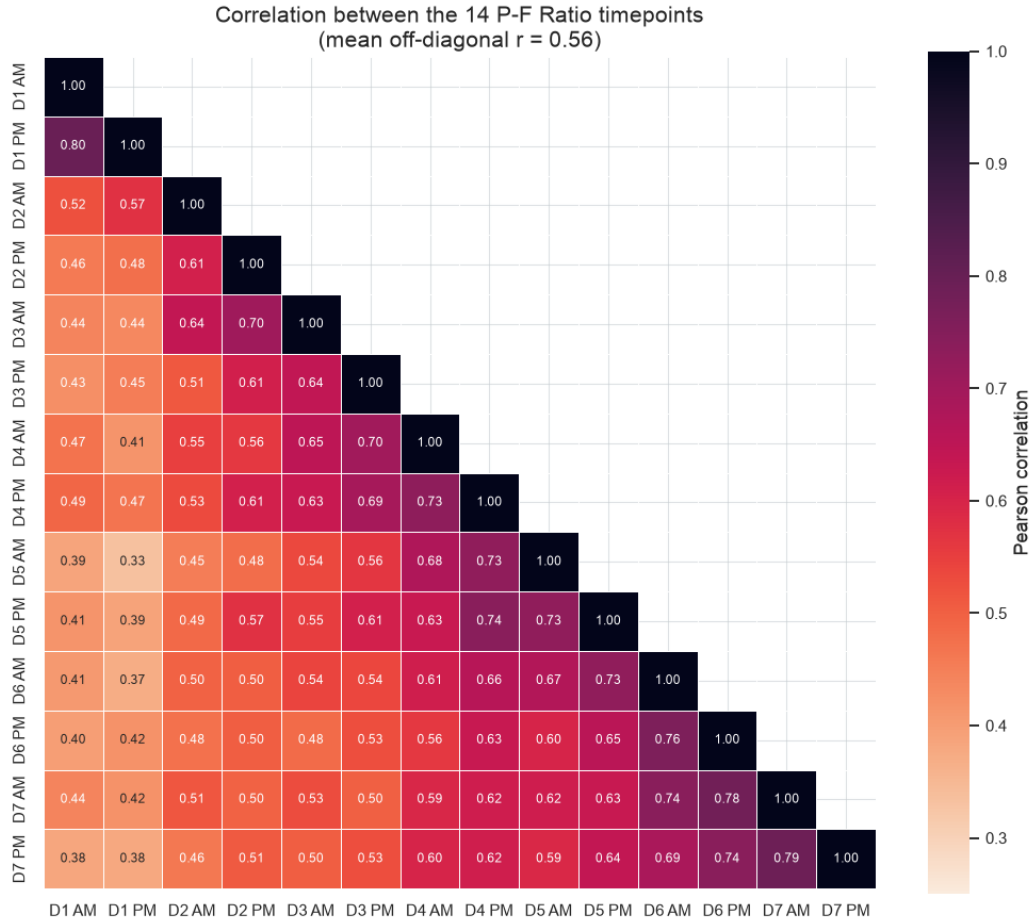


Figure 4: Pearson correlation between the fourteen timepoints, lower triangle only. The banded structure is what an autocorrelated series produces: adjacent readings correlate at $r = 0.71$, readings a week apart at $r = 0.38$.

Figure 4 makes visible the structure that a table of 196 numbers would hide. High values run along the diagonal and fade with distance, and a warmer block in the lower right shows the six timepoints from day-5 morning onward cohering more tightly than the rest. Besides the modelling consequences already described, this figure is what licenses the Benjamini–Hochberg correction in Section 3, since uniformly positive correlation is exactly the dependence structure under which that procedure remains valid.

Reducing each patient to a single point makes the trajectory visible, with the diagonal representing no change. Day one predicts day seven only moderately, $r = 0.45$, so the cloud is broad rather than hugging the diagonal, and the two centroids are displaced perpendicular to it as well as along it. That perpendicular component is new information, which is why the change score is significant on its own. Figure 5 equally shows why the engineered features did not help the classifiers: the clouds share one envelope and what moves is the centroid, not the population.

Figure 6 was the screening view used while working through Section 2, plotting raw value against patient index for each timepoint. Its value is partly negative: the isolated upward spikes appear at every timepoint and never twice for the same patient, which is consistent with occasional genuine extremes rather than a systematic recording fault and supports capping rather than deleting. Read against Section 1.2, the figure also shows that the measurements carry no visible structure along the index, so the ordering found in the outcome column is a property of that file alone rather than of the cohort, which is consistent with

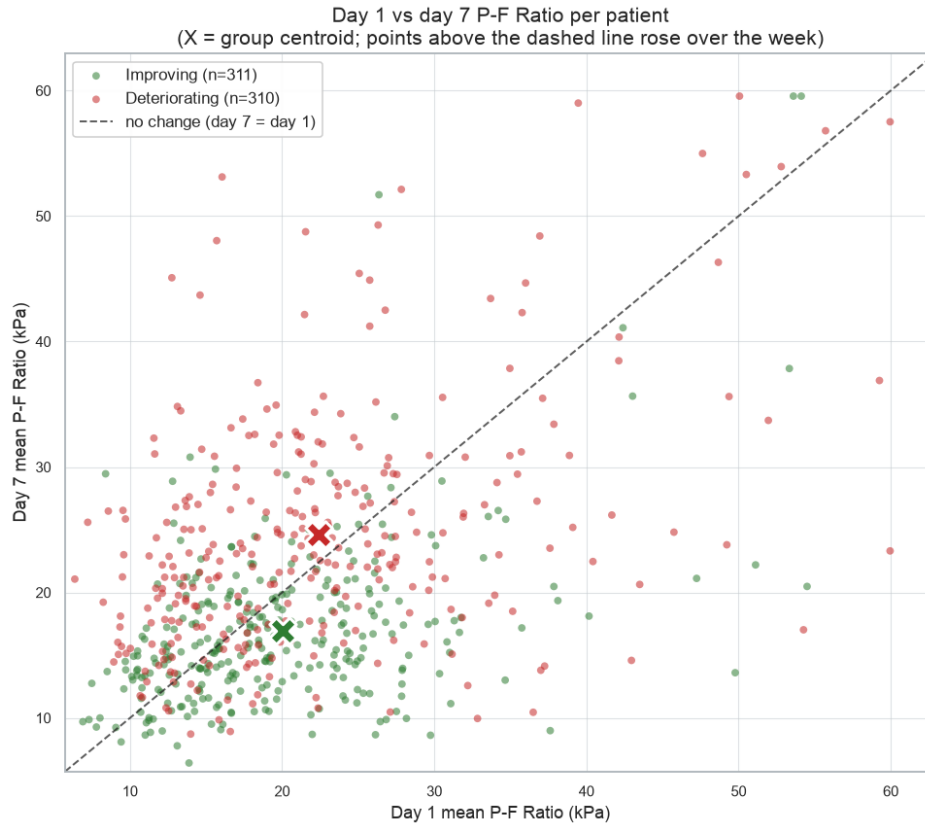


Figure 5: Day-1 against day-7 mean P/F ratio, one point per patient, group centroids marked. Points above the dashed line rose over the week: 62 % of *Deteriorating* patients against 35 % of *Improving* ones.

the changepoint being an artefact of how the label file was assembled.

5 Helper functions

The notebook defines twelve functions, of which seven are used more than once, and two of those carry most of the weight.

`flag_outliers(df, k, log)` began as `find_outliers`, which took one Series and returned a list of indices, so counting anything meant looping over fourteen columns and taking a union. Returning a boolean DataFrame instead makes rows, cells and per-column counts one line each, and exposing `k` and `log` as arguments is what made the five-strategy comparison in Table 2 affordable rather than five copies of the same code. Its companion `winsorise` answers the same rule by clipping instead of deleting, and both take their fences from one shared `_tukey_fences`, so they cannot disagree.

`evaluate_classifier` is what makes the model comparison fair. It fits the untouched default on the same training split, grid-searches over the same folds and scores on the same held-out test set. All three classifiers and all three sensitivity re-runs pass through it, so “default against tuned” and “model A against model B” compare what they claim to rather than six differently-built experiments.

The rest are smaller: `corrected_paired_t` for the Nadeau–Bengio correction, `mcnemar`, `short_labels` for axis labels derived from the column names, `summarise_by_outcome`, `compare_groups` for the per-timepoint test battery, and `plot_group_trajectory` used in Figures 1 and 12.

P-F Ratio by patient index, one panel per timepoint (raw data, before cleaning)

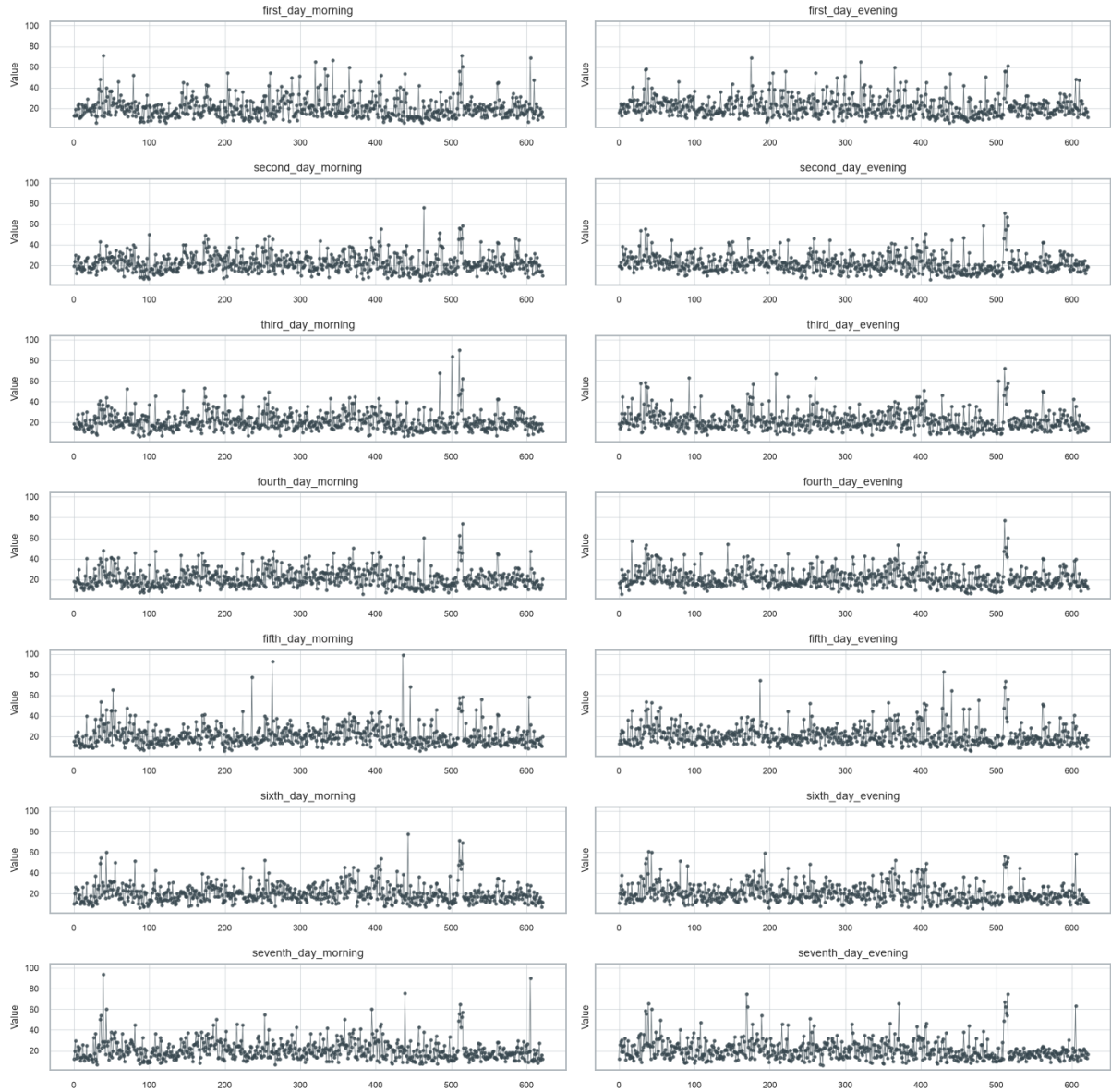


Figure 6: P/F ratio against patient index, one panel per timepoint, before cleaning. The measurements show no drift or blocking along the index, which is what makes the ordering found in the *outcome* column (Section 1.2) the more striking.

6 Classification

6.1 Setup

All three classifiers use one shared 80/20 stratified split with `random_state=42`, giving 496 training and 125 test patients. An 80/20 split was preferred to 70/30 because the dataset is small: a fifth still gives 125 test patients, enough for a usable estimate, while leaving as much data as possible for training and for the inner cross-validation the grid searches need. Scaling sits inside a Pipeline in every case, so the scaler is fitted on the training fold alone and never carries test statistics into training.

6.2 The three classifiers

For the logistic regression I reduced C from 1.0 to 0.01, a hundredfold increase in regularisation. Figure 4 established that the fourteen timepoints are heavily collinear, and under collinearity an under-regularised logistic regression distributes weight arbitrarily among correlated predictors and produces large, unstable, mutually cancelling coefficients. Figure 7 shows exactly that: at $C = 10$ the coefficients swing between -0.33 and $+0.20$ across adjacent, highly correlated timepoints, while at $C = 0.01$ they settle into a smooth profile rising towards the end of the week, which is what the effect sizes in Table 4 would lead one to expect. Accuracy barely moves between these solutions, while the model becomes far more interpretable, which is what regularisation is doing here. `max_iter` was also raised to 5000 so that `lbfgs` converges rather than stopping early.

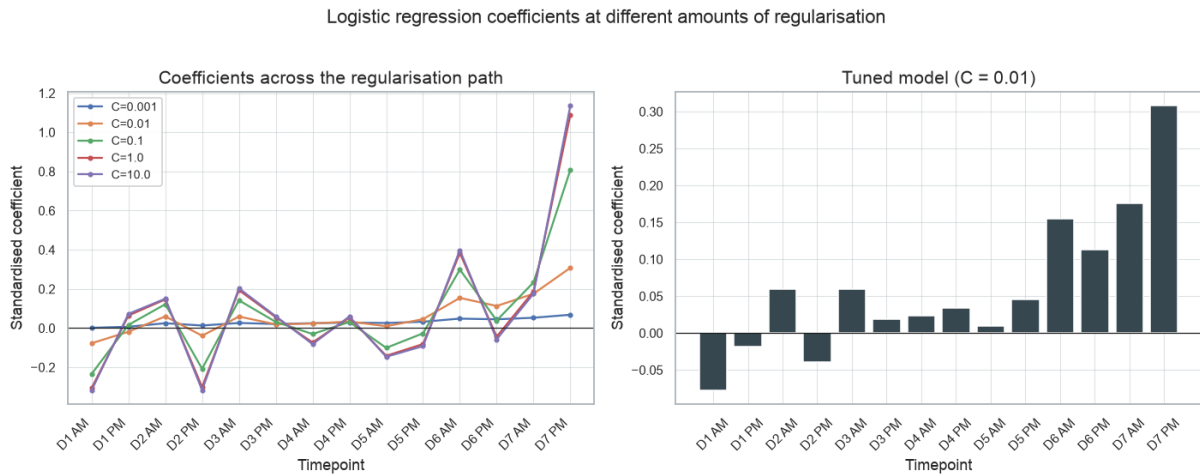


Figure 7: Standardised logistic-regression coefficients along the regularisation path (left) and for the tuned model at $C = 0.01$ (right). Weak regularisation makes adjacent, correlated timepoints oscillate; heavy regularisation produces a profile that rises towards day seven.

For the support vector classifier I raised C from 1.0 to 10.0 and `gamma` from 'scale' (about 0.07) to 0.5. These two jointly set the bias–variance trade-off of the RBF kernel: `gamma` controls how far one training point's influence reaches and C how hard the model tries to classify every training point correctly. Figure 3 showed heavily overlapping classes, so the boundary needs to be non-linear without chasing individual points through the overlap. They were searched jointly because they interact, and neither value means much alone. Figure 8 shows the whole grid spanning 0.623 to 0.762, a range of 0.139, which is wider than the gap between the best and worst of the three models.

The third classifier is a random forest rather than a Keras network, with `max_depth` constrained from None to 10 and `n_estimators` raised from 100 to 300. With 621 patients and fourteen features a neural network has far more parameters than the data can support and would mostly demonstrate overfitting, whereas a forest captures interactions between timepoints that logistic regression cannot, is unaffected by the residual skew because trees split on rank, and exposes feature importances that allow the Section 3 finding to be checked by a different route. Left at their defaults the trees grow until every leaf is pure, which

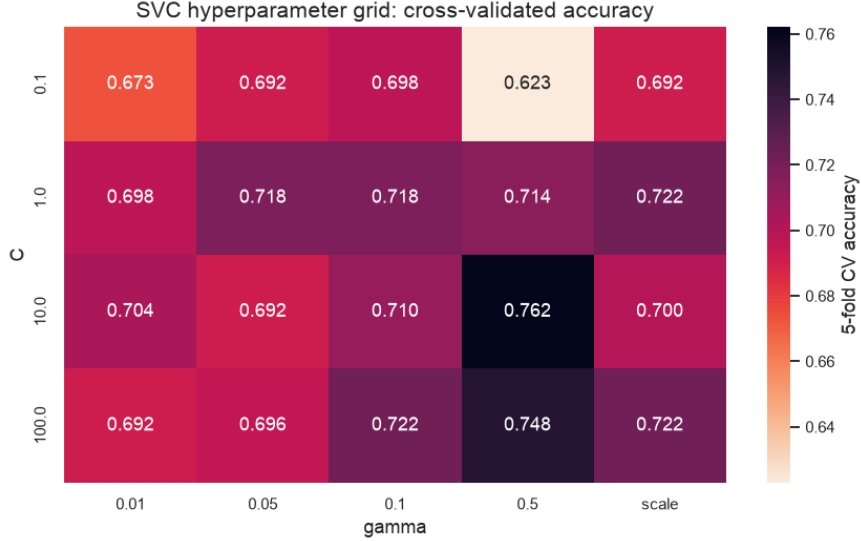


Figure 8: Cross-validated accuracy across the SVC hyperparameter grid. The spread of 0.139 exceeds the differences between the three classifiers, so choosing **gamma** badly costs more than choosing the wrong algorithm.

on 496 training patients means memorising the training set. The grid also searched `min_samples_leaf` but selected its default of 1, so the depth limit and the ensemble size are what actually changed.

Figure 9 is the return on that choice, with one honest qualification. The forest’s four most important timepoints are D7 PM, D7 AM, D6 PM and D6 AM; the four largest effect sizes are D7 PM, D7 AM, D6 AM and D6 PM. The same four, ranked by a non-parametric ensemble and by a parametric effect size. Across all fourteen, however, the rank correlation is $\rho = 0.486$ with $p = 0.078$, which is not significant, and the two measures are not independent since both are computed from the same 621 patients. The agreement on which timepoints matter most is therefore worth reporting, but it is corroboration rather than replication.

6.3 Results

Table 5 gives the required comparison. On the single held-out split the random forest scores highest at 0.792, ahead of the SVC at 0.776 and logistic regression at 0.736, against a majority-class baseline of 0.504. Figure 10 shows where those numbers come from: logistic regression misses 21 *Deteriorating* patients against the SVC’s 11, while the forest is the most evenly balanced of the three, and its ROC curve dominates over most of the range, giving the best AUC at 0.852.

Table 5: Classifier comparison. All three share one 80/20 stratified split. “Grid CV” is the best score found during tuning and is a maximum over the grid, so it is optimistically biased and the bias grows with grid size; it is reported for completeness but not used for ranking. “Nested CV” repeats the entire tuning procedure inside every outer fold and is the estimate the conclusions rest on.

Classifier	Accuracy (test)		Gain	Grid CV (biased)	Nested CV	ROC AUC
	default	tuned				
Logistic Regression	0.728	0.736	+0.008	0.698 ± 0.032	0.700 ± 0.031	0.809
SVC (RBF)	0.760	0.776	+0.016	0.762 ± 0.037	0.739 ± 0.052	0.806
Random Forest	0.776	0.792	+0.016	0.726 ± 0.041	0.725 ± 0.046	0.852
Majority-class baseline on the test set: 0.504. Grid sizes: 7, 20 and 20 candidates.						

Those test-set differences do not survive closer inspection, because the three models are scored on one shared test set, so their accuracies are paired rather than independent and McNemar’s exact test is the

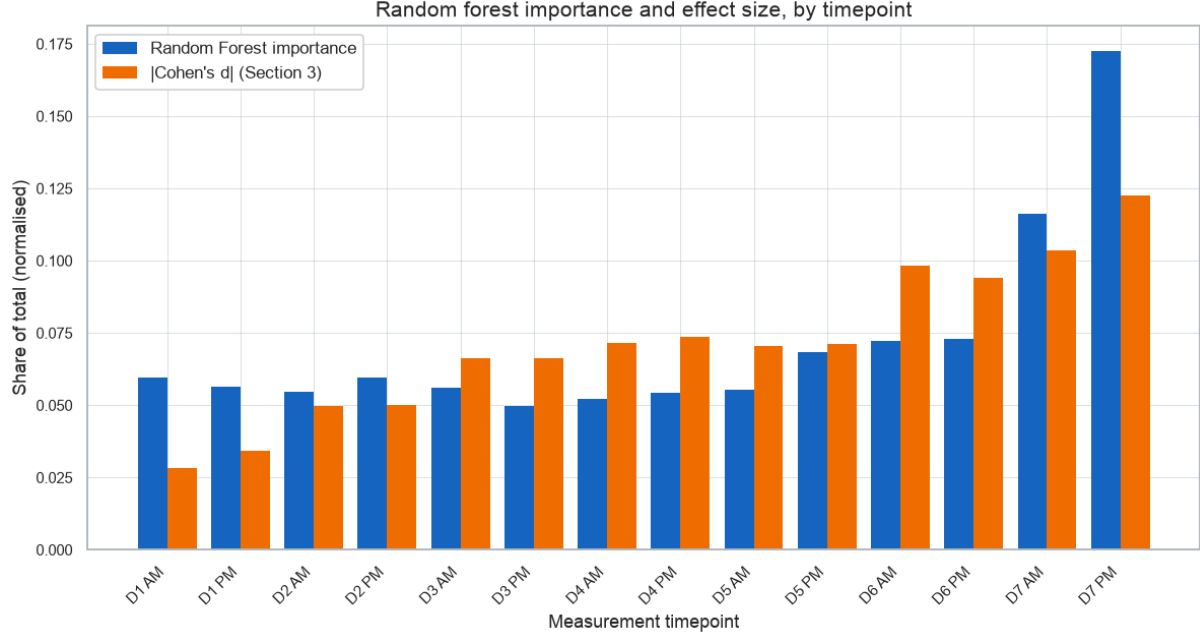


Figure 9: Random forest feature importances against the Cohen’s d values from Section 3, both normalised. The four highest-ranked timepoints are the same under both measures, though the rank correlation across all fourteen is not significant.

correct comparison. It gives $p = 0.442$ for the SVC against logistic regression, $p = 0.143$ for the forest against logistic regression and $p = 0.832$ for the forest against the SVC. Nothing separates them.

The resampled comparison had the same problem in a less obvious form, and correcting it changed the answer. Re-scoring an already-tuned model across the full dataset lets every test fold contain patients that were used to choose its hyperparameters. *The Elements of Statistical Learning* states the rule plainly: with a multistep modelling procedure, cross-validation must be applied to the entire sequence of steps, and samples must be left out before any selection. Figure 11 shows both estimates side by side. The optimism is 0.006 for logistic regression, 0.028 for the SVC and 0.009 for the forest. It is strongest for the SVC, whose selected ($C = 10$, $\gamma = 0.5$) sits on an isolated peak of its grid rather than a plateau, which is what happens when a search fits the folds rather than the problem. A larger search does not simply buy more optimism, at least not on three models: the SVC and the forest searched grids of equal size and differ threefold, and the forest is statistically indistinguishable from logistic regression despite a grid three times larger.

On the nested estimates the SVC leads at 0.739 ± 0.052 , ahead of the forest at 0.725 ± 0.046 and logistic regression at 0.700 ± 0.031 . Applying the Nadeau–Bengio correction, because repeated folds share training data, gives $p = 0.255$, $p = 0.223$ and $p = 0.595$ for the three pairwise differences. None is significant. Both the paired test on the test set and the corrected test on the resampled estimates reach the same verdict, which is the main empirical finding of this section.

6.4 Sensitivity to the cleaning decision

Repeating the whole comparison on the row-deleted dataset of 564 patients gives 0.735, 0.743 and 0.743 for logistic regression, the SVC and the forest, against 0.736, 0.776 and 0.792 under winsorisation. All three remain far above the baseline, all fourteen timepoints stay significant, and Cohen’s d at day-7 evening is essentially unchanged at +0.97 against +0.95 (Figure 12). The cleaning decision is not driving the results. The individual movements are informative in a different way: logistic regression shifts by 0.001, the SVC by 0.033 and the forest by 0.049, so the simplest model is also much the most stable under a change of preprocessing.

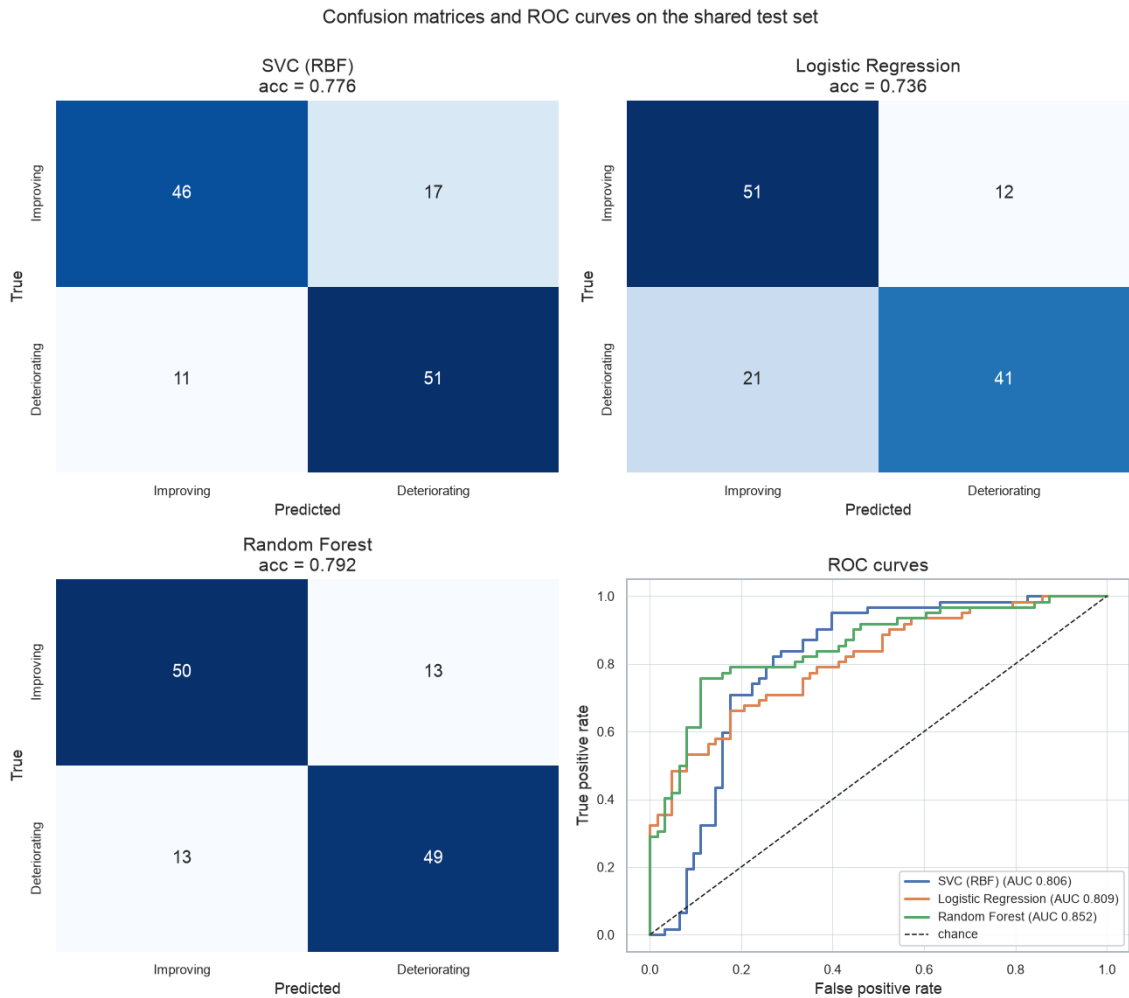


Figure 10: Confusion matrices for the three tuned classifiers on the shared 125-patient test set, with their ROC curves. No pair of classifiers differs significantly under McNemar’s test.

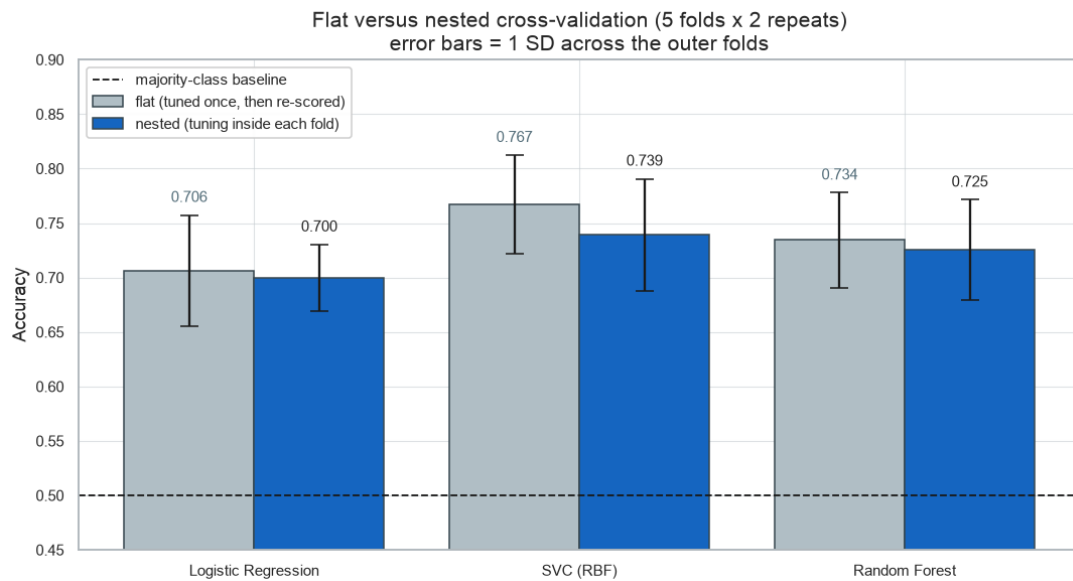


Figure 11: Flat against nested cross-validation, five folds by two repeats. The flat estimate re-scores hyperparameters that were chosen on the same data; the nested estimate refits the entire grid search inside every outer fold. The difference between the pairs is selection bias. It is largest for the SVC, whose chosen setting sits on an isolated peak of its grid.



Figure 12: Mean trajectories under the two cleaning rules. Winsorising and deleting the flagged rows produce the same picture, which is what the sensitivity analysis was run to establish.

7 Conclusions

7.1 Classifier comparison

Which classifier performed best? Is the difference meaningful or marginal?

On the evidence available none of the three performed best, and the differences between them are not meaningful. That conclusion is more useful than it first appears, because the obvious reading of Table 5 says otherwise: the random forest scores 0.792 on the held-out split and appears to win. That lead dissolves under two corrections. The three models share one test set, so McNemar's exact test applies and returns p between 0.14 and 0.83 for every pair. Redoing the resampled comparison with the tuning inside the loop moves the SVC down by 0.028 and reorders the top two, and the Nadeau–Bengio corrected differences are then $p = 0.255$, $p = 0.223$ and $p = 0.595$.

Cross-validation is estimating the expected error of a *procedure* across training sets, not the error of the particular model in hand, and those two quantities are only loosely related. The single test split is the only estimate aimed at the artefact itself, and on 125 patients it is far too noisy to rank models this close. Neither instrument can separate them, so the honest conclusion is that all three extract about the same amount of signal from this variable, roughly 0.70 to 0.74 against a 0.50 baseline, and the choice between them has to rest on something other than accuracy.

On that basis I would deploy the logistic regression. It is the most stable of the three under a change of preprocessing, moving 0.001 where the others move 0.033 and 0.049; it gives interpretable coefficients that can be shown to a clinician, and Figure 7 demonstrates those coefficients are stable once regularised; and it carries the smallest selection optimism, 0.006, so its reported number is the one least likely to disappoint in use. The case is not overwhelming, and there is a specific circumstance that would change it. An ROC curve is independent of class priors and misclassification costs, which is why it is the right curve to read when the operating point is not yet fixed. If this model were used for triage, with a threshold tuned to a chosen sensitivity rather than left at 0.5, the random forest's clearly better AUC of 0.852 against 0.809 would make it the better choice, because the quantity that matters then is how well patients are *ranked* rather than how many are correctly labelled at one arbitrary cut.

7.2 Hyperparameter tuning

What hyperparameter changes had the most impact? Did tuning improve results?

Tuning improved every model on the test set, by 0.008, 0.016 and 0.016, and every one of those gains is smaller than the noise in the estimate, so tuning did not measurably improve anything.

Where the hyperparameters mattered differed sharply between models, and that is the more interesting

result. For the SVC they mattered enormously: the grid in Figure 8 spans 0.139, so a poor choice of γ costs more than picking the wrong algorithm entirely. For logistic regression the effect on accuracy was negligible but the effect on the model was not. The search selected $C = 0.01$, a hundredfold more regularisation than the default, which barely moved the score while turning an oscillating set of coefficients into an interpretable one. For the random forest the constraints mattered least, since averaging over 300 trees already suppresses most of what limiting an individual tree would prevent.

The tuning also produced the clearest methodological result in the report, which is that the shortcut version of the comparison was measurably optimistic and that correcting it changed the ranking. Comparing the flat and nested estimates in Figure 11 makes the selection bias visible as a number rather than a warning. The lesson generalises past this dataset: a hyperparameter search is part of the model-fitting procedure, and any error estimate that does not contain it is measuring something else.

7.3 Further work

What would you do differently if you had more time or data?

The largest gain available would come from using more than one variable. All ten feature files hold 622 rows and share one `outcome.csv`, so they very probably describe the same patients, and Table 1 shows nine of the ten separate the classes on their own. Their per-patient means correlate at only 0.21 on average, an effective 8.0 independent tests out of ten, so they are very far from redundant, and a model given the P/F ratio together with CRP, peak pressure and lactate would almost certainly beat 0.74. That single change would likely matter more than every modelling decision in Section 6 combined.

The cleaning fences also belong inside the resampling loop rather than fitted once on everything; the leakage measured at 0.0002 here, but the correct construction costs little. The fourteen columns should be modelled as the time series they are rather than as independent features with a slope added afterwards. Patient-level covariates, meaning age, comorbidities and treatment received, are almost certainly stronger predictors than any single laboratory value, and without them the model is confounded in ways I cannot check from inside the dataset. And the evaluation should be reported in terms a clinician can act on, since accuracy weights a missed deterioration exactly as heavily as a false alarm.

Finally, the anomalies in Sections 1.1 and 1.2 would go back to whoever produced the data before anything built on it was used.

7.4 Predictability of the outcome

How well can patient outcomes be predicted from seven days of measurements?

A single variable predicts the outcome better than might be expected, and nowhere near well enough to use on its own. Roughly 0.74 accuracy under nested cross-validation, with an AUC between 0.81 and 0.85, from fourteen readings of a single laboratory value against a 0.50 baseline, is a real and reproducible signal. It survived a change of cleaning rule, three different algorithms and a correction that removed the selection bias from the estimate. As one component of a risk score it would be useful.

Three things bound how much further it could go, and only one of them is a modelling problem. The classes overlap: at the most discriminative timepoint the *Deteriorating* median sits at the *Improving* 91st percentile, which is strong separation, but the distributions still share most of their range, and a large minority of patients fall where no boundary can be right for both. This is a property of the data rather than of any model, which is why three very different algorithms converge on the same narrow band. The signal also arrives late, $d = 0.22$ on admission against 0.95 on day seven, so a model of this kind becomes reliable at roughly the point where the outcome is becoming clinically apparent anyway, which limits its value for the decision a clinician actually faces on day one. And a single physiological axis cannot represent a multi-organ disease; the ceiling here is set by what one variable can know.

My own reading is that seven days of a single measurement can stratify patients into higher- and lower-risk groups usefully, but cannot predict an individual outcome to a standard that would justify a treatment decision. The defensible framing is a triage aid that directs attention, not a prediction. And

nothing built on this dataset should be deployed until the label direction and the row ordering documented in Section 1 have been explained.

Annex: Declaration of Generative AI Use

Student Name: Barbora Šandová

Student ID (BT Code): 731058

Date: 30 August 2026

1. Disclosure confirmation

I confirm that every section of code in my submitted notebook that was generated or substantially assisted by a generative AI tool has been enclosed within the required AI ASSISTED SECTION markers, as specified in the coursework instructions. No AI-assisted work has been left undisclosed.

2. AI tools used

Claude Opus 5 (via Claude Code), used for drafting and refactoring code, for wording and summarising analysis that I specified, and for adversarial review of my own results.

3. Estimated contribution

Of the code and written analysis in my submission, I estimate the following breakdown:

Written by me (human-authored)	60 %
Generated or substantially assisted by AI	40 %

The knowledge and the judgement behind this submission are my own. I directed the tool throughout: I decided what to analyse, chose the methods, reviewed every result and rejected what did not hold up. Nothing here was generated without my supervision, and I can explain and defend every part of it.

Declaration

I confirm that the information provided above is accurate and complete. I understand that submitting work that I do not understand or cannot explain is a violation of academic integrity.

Signature: _____

Date: 30 August 2026